

Sungsoo Han

Master's Degree

Department of Electrical and Information Engineering

Seoul National University of Science and Technology, Seoul, Republic of Korea

Homepage: <https://hansungsoo98.github.io> ; E-mail: hanss9805@seoultech.ac.kr

Google Scholar: <https://scholar.google.com/citations?user=AvOe5dEAAAAJ&hl=ko>

EDUCATION

Aug. 2024	Seoul National University of Science and Technology	Electrical and Information Engineering	M. S.
Aug. 2023	Seoul National University of Science and Technology	Electrical and Information Engineering	B. S.

EXPERIENCES

Seoul National University of Science and Technology, SeoulTech	Seoul, Korea
Research intern / Advisor: Hyun Kim	Jan. 2024 - Sep. 2024
Korea Institute of Science and Technology, KIST	Seoul, Korea
Research intern / Advisor: Jaewook Kim	Sep. 2022 - Sep. 2023

RESEARCH INTERESTS

- ♦ **Efficient AI Computing: Model compression and accelerator architectures for efficient DNN and LLM inference**
- ♦ **Low-Precision Data Representation: Block quantization, logarithmic number systems, and outlier-aware data formats**
- ♦ **Hardware-Software Co-Design: Hardware-aware algorithm development and RTL-based accelerator design**

INTERNATIONAL JOURNALS

1. **Sungsoo Han**, Dahun Choi, and Hyun Kim, "BGM: A Block Processing Unit-Integrated GEMM Accelerator for Hardware-Efficient LLM Inference via Outlier-Aware Merge-and-Split Block Quantization," *IEEE Transactions on Circuits and Systems I*.
2. NamJoon Kim, Beom Jin Kang, Hae In Lee, Seokkyu Yoon, Youngchan Kim, **Sungsoo Han**, Seok-Bum Ko, and Hyun Kim, "FLASH: Energy-Efficient FPGA Acceleration via Linear Approximation and Streamlined Two-Stage Pipeline Architectures for Quantized CNN-Transformer Hybrid Networks", *IEEE Transactions on Circuits and Systems for Video Technology*.
3. Youngchan Kim*, Seokkyu Yoon*, **Sungsoo Han**, Chaewon Park, Junoh Park, Junha Go, and Hyun Kim, "Accelerating Language Giants: A Survey of Optimization Strategies for LLM Inference on Hardware Platforms", *Journal of Systems Architecture (JSA)*.

INTERNATIONAL CONFERENCES

1. **Sungsoo Han**, Dahun Choi, and Hyun Kim, "BOLD-Q: Blockwise Outlier-aware Logarithmic Dual-Bias Quantization for Hardware-Efficient LLM Inference," *Design, Automation and Test in Europe Conference 2026 (DATE 2026)*. (IF: 2, NRF BK21+).
2. **Sungsoo Han**, Dahun Choi, and Hyun Kim, "Closit: Clipped-Regime Posit Quantization for Edge-Friendly Vision Transformers," *IEEE Asia Pacific Conference on Circuits and Systems (APCCAS)*, Busan, South Korea, Oct. 12-15, 2025.

DOMESTIC CONFERENCES

1. **한성수**, 최다훈, **김현**, "저정밀도 비전 트랜스포머 추론을 위한 Additive Power-of-Two 융합 Microscaling 양자화 방법," 2025 년도 대한전자공학회 하계종합학술대회, Jun. 2025.
2. 이준식, 황인성, **한성수**, 김현, "하드웨어 자원 제약을 고려한 최적화된 FSRCNN 모델 자동 탐색 시스템," 2025 년도 대한전자공학회 하계종합학술대회, Jun. 2025.
3. 이정호, 정상범, **한성수**, 김현, "IQR 기반 이상치 탐지를 활용한 FSRCNN 양자화 기법," 2025 년도 대한전자공학회 하계종합학술대회, Jun. 2025.
4. **Sungsoo Han**, Dahun Choi, and **Hyun Kim**, "An Efficient Computing Unit Integrating Floating Point and Posit for Transformer Accelerators," 제 32 회 한국반도체학술대회 (KCS 2025), Pages: 933, Feb. 2025. (**현장우수포스터상**)
5. **한성수**, 최다훈, **김현**, "Local-Outlier Factor 를 활용한 Vision Transformer 의 신뢰도 높은 Clipping Value 탐색 기법," 제 7 회 반도체공학회 동계종합학술대회, Pages: 13-14, Jan. 2025. (**우수논문상**)
6. 강희범, 최다훈, **한성수**, 김현, "Low Rank 근사화 기법을 활용한 Vision Transformer 경량화 기법," 2024 년도 대한전자공학회 하계종합학술대회, Pages : 1865-1867, Jun. 2024. (**우수학생논문상**)

AWARDS

1. [2025.10.19] **한성수**(석사과정), Micro2025 AI Model Benchmarking Competition "Bronze Award" 수상
2. [2025.02.14] **한성수**(석사과정), 제 32 회 한국반도체학술대회 (KCS 2025) "현장우수포스터상" 수상
3. [2025.01.13] **한성수**(석사과정), 제 7 회 반도체공학회 동계종합학술대회 우수논문상 수상

IP (PATENT / SW)

1. 김현, **한성수**, 최다훈, "온디바이스 대규모 언어 모델 추론을 위한 로그 수 체계 기반 이중 바이어스 양자화 프레임워크 및 이를 이용한 연산 시스템 및 방법," 한국, 서울과학기술대학교 산학협력단, 10-2025-0212917, 2025 년 12 월 29 일(출원).
2. Hyun Kim, Sungsoo Han, Dahun Choi, "Logarithmic Number System-Based Dual-Bias Quantization Framework for On-Device Large Language Model Inference and System and Method Using the Same," 미국, 서울과학기술대학교 산학협력단, 19/665674, 2026 년 5 월 1 일(출원).
3. 김현, **한성수**, 최다훈, "하드웨어 효율적인 클리핑 기반 Closit(클로짓) 포맷 양자화 프로그램," 서울과학기술대학교 산학협력단, C-2025-057438, 2025 년 12 월 15 일(등록).

RESEARCH PROJECTS

2022-2024	Development of Low Power AI Architecture for AIoT, 산업통상자원부 지원
2024-2026	Development of AI Deep-Learning Processor and Module for 2,000 TFLOPS Server, 과학기술정보통신부 지원
2025-Present	Development of Transformer Acceleration Solution Including Quantization and Scheduling Techniques for LLM, 산업통상자원부 지원

SKILLS

EDA Tools: Design Compiler, Xilinx Vivado; Programming language: C/C++, Python; Hardware Description language: Verilog

REFERENCES

Prof. Hyun Kim	Associate Professor, Electrical and Information Engineering, Seoul National University of Science and Technology
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